MediScan Ai

Research Report I

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2022-AG-7688

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ABSTRACT

Healthcare systems worldwide are generating a massive volume of medical data in the form of textual clinical reports and diagnostic medical images such as X-rays and CT scans. Manual analysis of this data by healthcare professionals is time-consuming, resource-intensive, and prone to human error, which may affect diagnostic accuracy and patient outcomes. With the increasing demand for fast and reliable medical decision support, there is a strong need for intelligent systems that can assist medical professionals in analyzing and interpreting complex medical data efficiently.

The proposed Final Year Project, **MediScan AI**, aims to develop an artificial intelligence-based diagnostic assistance system that integrates **Natural Language Processing (NLP)** and **Computer Vision (CV)** techniques. The system will automatically extract key medical information from unstructured textual medical reports using NLP techniques such as Named Entity Recognition and transformer-based models. In parallel, computer vision models based on Convolutional Neural Networks will analyze medical images to detect abnormalities such as pneumonia, fractures, and tumors.

Both modules will be integrated into a unified web-based platform where healthcare professionals can upload medical reports or images and receive structured summaries along with predictive insights. The system will be evaluated for accuracy, reliability, and explainability to ensure practical usability. MediScan AI is expected to reduce diagnostic time, assist medical professionals in decision-making, and improve the overall efficiency and quality of healthcare services.

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Chapter 1 – INTRODUCTION

Background

The healthcare industry has experienced rapid digital transformation in recent years, leading to the generation of vast amounts of medical data. This data includes electronic health records, diagnostic reports, and medical imaging data such as X-rays and CT scans. Despite advancements in healthcare technology, a significant portion of medical data remains unstructured, making it difficult to analyze efficiently. Medical professionals often rely on manual interpretation of reports and images, which is time-consuming and susceptible to human error.

Artificial Intelligence has emerged as a powerful tool to address these challenges by automating data analysis and supporting clinical decision-making. Natural Language Processing enables machines to understand and extract meaningful information from textual data, while Computer Vision allows automated analysis of medical images. Combining these technologies can significantly improve diagnostic accuracy and reduce workload on healthcare professionals.

MediScan AI is proposed to bridge this gap by providing an intelligent diagnostic assistant that processes both textual and visual medical data. The system aims to support healthcare professionals by offering quick insights, reducing diagnostic delays, and improving patient care outcomes.

1.2 Description

MediScan AI is an AI-powered healthcare diagnostic support system designed to analyze medical reports and diagnostic images using advanced machine learning techniques. The system focuses on two primary data sources commonly used in healthcare: textual clinical reports and medical images such as X-rays.

The NLP module will process unstructured medical reports to extract key medical entities such as diseases, symptoms, medications, and clinical findings. Transformer-based models like BioBERT and ClinicalBERT will be utilized to enhance medical language understanding. The extracted information will be structured in a readable format for healthcare professionals.

The Computer Vision module will analyze medical images using Convolutional Neural Networks to identify abnormalities such as pneumonia, fractures, or tumors. Pretrained deep learning models will be fine-tuned on labeled medical datasets to achieve high diagnostic accuracy.

Both modules will be integrated into a web-based platform that allows users to upload reports or images easily. The backend system will process the input and return predictions along with structured summaries. The platform aims to be user-friendly, reliable, and secure, making it suitable for real-world healthcare environments.

1.3 Problem Statement

Healthcare professionals face challenges in analyzing large volumes of unstructured medical data efficiently. Manual interpretation of medical reports and diagnostic images is time-consuming, prone to errors, and may lead to delayed or inaccurate diagnoses. There is a lack of integrated systems that can simultaneously analyze textual and visual medical data to assist clinicians in decision-making.

1.4 Scope

The scope of MediScan AI includes the development of an AI-based diagnostic assistant capable of analyzing medical reports and X-ray images. The system will support disease detection and information extraction using NLP and Computer Vision techniques. It will provide predictive insights through a web-based interface. The project focuses on assistance rather than replacing medical professionals and will be limited to predefined datasets and supported medical conditions.

1.5 Objectives

Goal:
To develop an AI-powered diagnostic assistance system for healthcare professionals.

Objectives:

- To extract key medical information from textual medical reports using NLP.
- To detect abnormalities in medical images using Computer Vision.
- To integrate NLP and CV modules into a single web-based platform.
- To ensure high accuracy, reliability, and explainability of predictions.

1.6 Feasibility

1.6.1 Technical Feasibility

The project utilizes well-established AI frameworks such as TensorFlow and PyTorch, making it technically feasible.

1.6.2 Schedule Feasibility

The project can be completed within the academic timeline using an Agile development approach.

1.6.3 Economic Feasibility

Open-source tools and free-tier cloud services will minimize development costs.

1.6.4 Cultural Feasibility

The system supports healthcare improvement without cultural or societal conflicts.

1.6.5 Legal/Ethical Feasibility

The project uses publicly available datasets and follows ethical AI practices.

1.6.6 Resource Feasibility

Required computational and human resources are available.

1.6.7 Operational Feasibility

The system is easy to operate and supports healthcare workflows.

1.7 Requirements

This section defines the functional and non-functional requirements necessary for the successful development and operation of the MediScan AI system. The requirements ensure that the system meets user expectations, performs reliably, and complies with academic and practical standards.

1.7.1 Functional Requirements

FR ID	Requirement Name	Description
FR01	User Authentication	System shall allow users to register and log in using valid credentials.
FR01-01	User Registration	System shall allow new users to create an account.
FR01-02	User Login	System shall authenticate user credentials before granting access.
FR01-03	Invalid Login Handling	System shall display an error message for invalid login attempts.
FR02	Upload Medical Reports	System shall allow users to upload textual medical reports.
FR02-01	Report Format Validation	System shall support PDF and text-based medical reports.
FR02-02	Secure Report Upload	System shall securely upload and store medical reports.
FR03	Upload Medical Images	System shall allow users to upload diagnostic medical images.
FR03-01	Image Format Support	System shall support image formats such as JPG and PNG.
FR03-02	Image Validation	System shall validate medical images before analysis.
FR04	NLP-Based Report Analysis	System shall analyze medical reports using NLP techniques.
FR04-01	Entity Extraction	System shall extract diseases, symptoms, medications, and findings.
FR04-02	Report Summarization	System shall generate a structured summary of extracted data.
FR05	Computer Vision Analysis	System shall analyze medical images using deep learning models.

FR05-01	Abnormality Detection	System shall detect abnormalities such as pneumonia or fractures.
FR05-02	Prediction Generation	System shall generate predictive results from image analysis.
FR06	Result Presentation	System shall display analysis results to the user.
FR06-01	Confidence Scores	System shall provide confidence or probability scores with results.
FR07	Data Management	System shall store analysis results securely for future access.
FR07-01	View Previous Results	Authorized users shall view previously analyzed reports and images.

1.7.2 Non-Functional Requirements

NFR01: Availability

- The system shall be available 24/7 with minimal downtime.

NFR02: Performance

- The system shall process reports and images within an acceptable response time.

NFR03: Usability

- The system shall provide a user-friendly and intuitive interface.
- The interface shall be accessible to users with basic computer literacy.

NFR04: Security

- The system shall ensure confidentiality and privacy of medical data.
- User data shall be protected using authentication and secure storage mechanisms.

NFR05: Scalability

- The system shall be scalable to handle an increasing number of users and data inputs.

NFR06: Maintainability

- The system shall be modular and well-documented to support future enhancements.

1.7.3 Hardware Requirements

- **Processor:** Intel Core i3 or above
- **RAM:** Minimum 4 GB (8 GB recommended)
- **Storage:** 40 GB or more
- **Internet Connection:** Required for web-based access

1.7.4 Software Requirements

- **Operating System:** Windows 10 or above / Linux
- **Web Browser:** Google Chrome, Mozilla Firefox
- **Backend Language:** Python
- **Frontend Technologies:** HTML, CSS, JavaScript
- **Frameworks & Libraries:** TensorFlow, PyTorch, Flask/FastAPI

1.8 Stakeholders

The following stakeholders are involved in or affected by the MediScan AI system:

- **Healthcare Professionals:** Primary users who benefit from diagnostic assistance
- **Patients:** Indirect beneficiaries through improved diagnostic efficiency
- **System Administrators:** Responsible for system maintenance and management
- **Project Supervisor:** Provides academic guidance and evaluation

Chapter 2 – MATERIALS & METHODS

2.1 Process Model

The Agile development model is adopted for the MediScan AI project. This model supports iterative development, continuous testing, and frequent feedback, which are essential for AI-based systems. The project is divided into multiple sprints, each focusing on specific modules such as data preprocessing, NLP model development, image analysis, and system integration. This approach allows flexibility, timely improvements, and efficient handling of evolving project requirements.

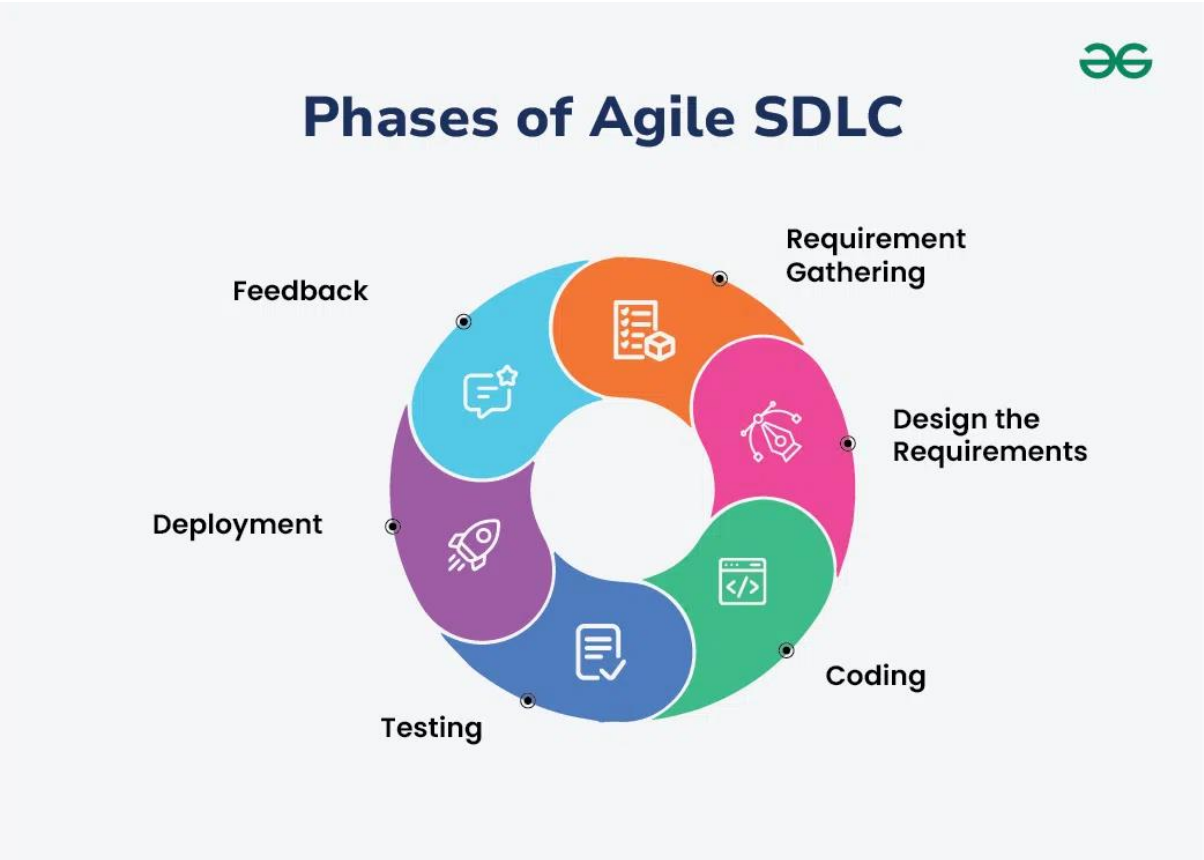

Figure 1. Agile Development Model for MediScan AI

2.2 Tools & Technologies

The following tools and technologies are used in the development of MediScan AI:

- **Python:** Core programming language for system development
- **TensorFlow & PyTorch:** Used for building and training deep learning models
- **HuggingFace Transformers:** Used for NLP tasks and medical text processing
- **OpenCV:** Used for medical image preprocessing
- **Flask / FastAPI:** Backend framework for system APIs
- **HTML, CSS, JavaScript:** Frontend interface development
- **Git & GitHub:** Version control and collaboration

Chapter 3 – TIMELINE

A Gantt chart is used to represent the project timeline. The project phases include requirement analysis, dataset collection, model development, system integration, testing, documentation, and final submission. The timeline ensures that all activities are completed within the academic schedule.

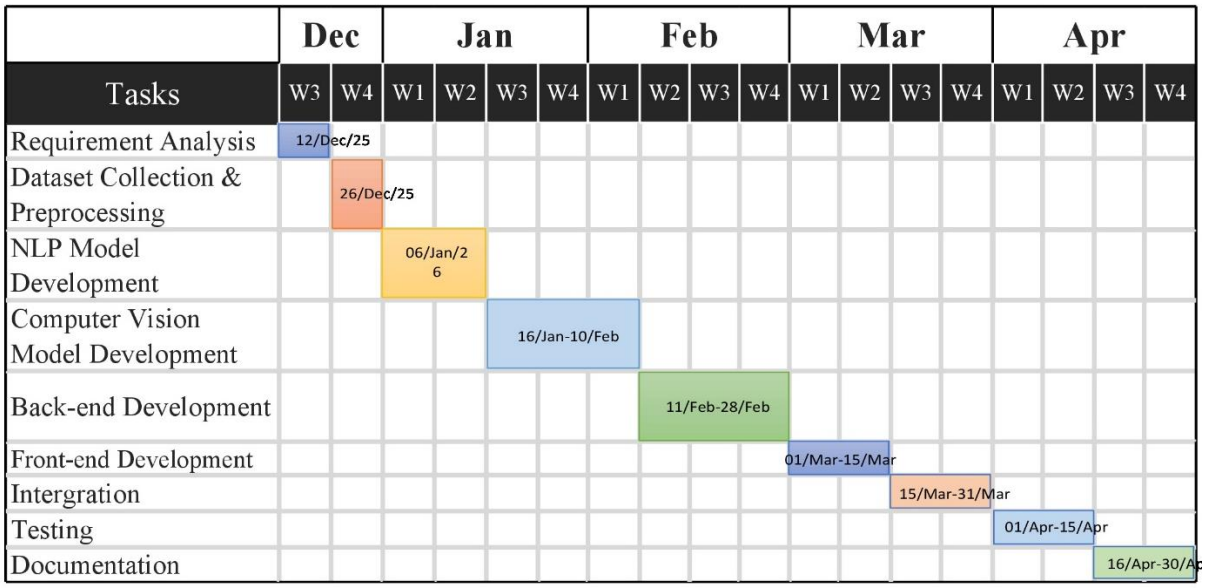

Figure 2. Project Timeline (Gantt Chart)